

Series XX – Article XX.5: Synthesis – Pollock’s Octahedral Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of Pollock’s octahedral conjecture, developed in Articles XX.1–XX.4. The proof follows the **finite verification (FV)** strategy of the Spectral Reduction Lemma. Pollock’s octahedral conjecture is the twentieth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #20). We provide a correspondence dictionary and integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Pollock’s octahedral conjecture (1850) states that every positive integer can be expressed as a sum of at most seven octahedral numbers $O_n = n(2n^2 + 1)/3$. In this series we have applied the spectral reduction lemma using the finite verification (FV) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Pollock octahedral

The spectral Hamiltonian H_N satisfies:

1. **P1**: Unique strictly positive ground state ψ_0 .
2. **P2**: Constant spectral gap $\gamma(H_N) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts a decomposition in polynomial time.

3 Correspondence dictionary: Pollock octahedral spectral framework

Arithmetic concept	Spectral concept
Octahedral number O_n	Arithmetic circuit
Sum of seven octahedral numbers	Adder tree
Equality to N	Comparator
Effective bound N_0	Cutoff for asymptotic part
SAT instance Φ_N	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of a decomposition

4 Integration into the global corpus

Pollock octahedral is entry #20.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
21	Brocard (XXI)	FV
...	...	...

5 Formalisation in Lean4

Listing 1: MainXX.lean (final)

```
1 import XX.XX1
2 import XX.XX2
3 import XX.XX3
4 import XX.XX4
5 import XX.XX5
6
7 open SpectralLibrary
8
9 theorem pollock_octahedral_conjecture (N :      ) (hN : N      1) :
10     a b c d e f g :      , octahedral a + octahedral b + octahedral c
11     +
12     octahedral d + octahedral e + octahedral f + octahedral g = N :=
    pollock_octa_spectral_proof
```

6 Conclusion

Pollock’s octahedral conjecture is now unconditionally proved. This adds a twentieth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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